

Loan Approval Prediction & Process Improvement

Business Requirements Document (BRD)

Improving loan approval decisioning through a risk-tiered, evidence-based process

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1. Document Control

1.1 Version History

Version	Date	Author	Description of Change
0.1	11-07-2024	Chirag Seervi	Initial draft based on SQL/Python analysis findings
1.0	23-07-2024	Chirag Seervi	Finalized with stakeholder-ready recommendations

1.2 Approvers / Sign-off

Name	Role	Sign-off Date
[Head of Lending]	Sponsor	[Date]
[Risk/Credit Manager]	Business Stakeholder	[Date]
[Ops Manager]	Process Owner	[Date]

2. Executive Summary

Analysis of historical loan applicant data shows that approval decisions are driven overwhelmingly by a small set of factors — most notably credit history, income level, and requested loan amount relative to income. Applicants without a credit history are approved only ~18% of the time versus ~79% for those with one, yet the current process does not consistently separate these applicants for differentiated handling. This BRD proposes a risk-tiered approval workflow that routes low-risk applicants through a streamlined path and flags high-risk applicants (missing credit history, low income, high loan amount) for manual review rather than uniform automatic processing.

3. Business Problem / Opportunity

3.1 Problem Statement

The lending team lacks a structured, evidence-based way to differentiate applicant risk at the point of approval, leading to inconsistent decisioning — some low-risk applicants face unnecessary delay, while some high-risk applicants may be approved without adequate scrutiny.

3.2 Business Impact if Unaddressed

- Elevated default risk from high-risk applicants (missing credit history, low income, high loan amount) approved without enhanced review.
- Lost time-to-decision and applicant experience cost for low-risk applicants routed through the same lengthy process as high-risk ones.
- No systematic triage criteria, leaving approval decisions dependent on individual reviewer judgment rather than consistent data-backed rules.

3.3 Root Cause (from analysis)

SQL-based analysis of the applicant dataset identified credit history as the single strongest predictor of approval (79% vs. 18%), followed by income level (+16% approval likelihood) and loan amount relative to income (low-risk profiles show a 62% higher approval rate than high-risk profiles). The current process does not formally weight or triage on these factors.

4. Business Objectives & Success Metrics

Objective	Success Metric (KPI)	Target	Baseline
Reduce manual review time for low-risk applicants	Avg. time-to-decision, low-risk tier	[e.g. -30%]	[Current avg.]
Reduce default risk from high-risk approvals	Default rate on flagged-but-approved loans	[e.g. -10%]	[Current rate]
Increase decision consistency	% of decisions following triage criteria	[e.g. 90%+]	[Baseline unknown]

5. Stakeholders

Name / Team	Role	Interest / Stake	RACI
[Head of Lending]	Sponsor	Owns approval outcomes & risk exposure	A
[Credit/Risk Team]	Subject Matter Expert	Defines risk scoring rules	R
[Loan Officers]	End Users	Execute triage & manual reviews	R
[Compliance]	Reviewer	Ensures fair-lending compliance	C
[IT/Dev Team]	Build Team	Implements scoring logic in system	R
[Applicants]	Indirect Stakeholder	Experience faster/fairer decisions	I

6. Scope

6.1 In Scope

- Defining risk-tier triage criteria based on credit history, income, and loan amount.
- Designing a streamlined approval path for low-risk applicants.
- Designing an enhanced manual-review path for high-risk applicants (2+ risk flags).
- Stakeholder dashboard to monitor approval patterns and rule performance.

6.2 Out of Scope

- Changes to core credit scoring/bureau integration (assumed to remain as-is).
- New loan products or pricing changes.

7. Assumptions, Constraints & Dependencies

7.1 Assumptions

- Historical applicant data used for analysis reflects current applicant population characteristics.
- Credit history field is reliably populated at time of application.

7.2 Constraints

- Any automated triage must remain compliant with fair-lending regulations (no proxy discrimination via correlated variables such as gender or location).

7.3 Dependencies

- Requires sign-off from Compliance before automated triage rules go live.
- Requires IT to implement scoring/triage logic in the loan origination system.

8. Current State vs. Future State (As-Is / To-Be)

See BPMN process diagram (linked separately). Summary below:

As-Is Summary: All applications, regardless of risk profile, follow the same manual review sequence — credit check, income verification, manual underwriter review, decision. No differentiated path by risk tier.

To-Be Summary: Applications are triaged at intake: low-risk profiles (credit history present, adequate income, low loan amount relative to income) route through a lightweight approval path; applicants with 2+ risk flags (no credit history, low income, high loan amount) route to enhanced manual review.

9. Risks

Risk	Likelihood	Impact	Mitigation
Triage rules inadvertently correlate with protected attributes	Medium	High	Compliance review of rule variables before go-live; exclude gender/location from scoring logic
Streamlined path approves risk that manual review would have caught	Low	Medium	Phased rollout with monitoring dashboard; review approval outcomes monthly for first quarter

10. High-Level Timeline

Phase	Milestone	Target Date
Discovery	Requirements & rule definition sign-off	[Date]
Design	FRD & triage logic approved	[Date]
Build	Triage logic implemented in system	[Date]
UAT	Rule testing & stakeholder sign-off	[Date]
Launch	Phased rollout begins	[Date]

11. Related Documents

- Functional Requirements Document (FRD) — Loan Approval Risk-Tiered Process (companion document)
- Source analysis: SQL-based approval/rejection driver analysis (Python cleaning + EDA, SQL business analysis)
- Stakeholder dashboard, BPMN process diagram, Jira backlog